Date: ________________

Lab # 03 TO SIMPLIFY COMPLEX LOGIC CIRCUITS USING DE MORGAN'S THEOREM & BOOLEAN ALGEBRA IDENTITIES

OBJECTIVES:

O1: To study De Morgan's theorem, Boolean algebra identities and its applications.

O2: To check the validity of De Morgan's theorem by using it to simplify a complex logic circuit.

BACKGROUND THEORY

In propositional logic and Boolean algebra, De Morgan's laws are a pair of transformation rules that are both valid rules of inference. The rules allow the expression of conjunctions and disjunctions purely in terms of each other via negation. The rules can be expressed in English as:

"THE NEGATION OF A CONJUNCTION IS THE DISJUNCTION OF THE NEGATIONS."

AND

"THE NEGATION OF A DISJUNCTION IS THE CONJUNCTION OF THE NEGATIONS."

In Boolean expressions:

$$\overline{(\boldsymbol{A}.\boldsymbol{B})} = \overline{\boldsymbol{A}} + \overline{\boldsymbol{B}}$$

AND

$$\overline{(\boldsymbol{A}+\boldsymbol{B})} = \overline{\boldsymbol{A}}.\overline{\boldsymbol{B}}$$

Applications of the rules include simplification of logical expressions in computer programs and digital circuit designs. DeMorgan's theorem may be thought of in terms of breaking a long bar symbol. When a long bar is broken, the operation directly underneath the break changes from addition to multiplication, or vice versa, and the broken bar pieces remain over the individual variables. To illustrate:

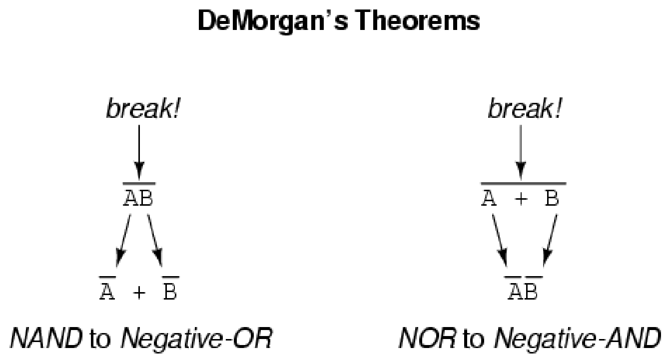

Fig 3.1: Illustration of De Morgan's Theorem

EXAMPLE CONVERSION OF COMPLEX LOGIC FUNCTION TO SIMPLE LOGIC FUNCTION

De Morgan's theorem may be thought of in terms of breaking a long bar symbol. When a long bar is broken, the operation directly underneath the break changes from addition to multiplication, or vice versa, and the broken bar pieces remain over the individual variables. When multiple "layers" of bars exist in an expression, you may only break one bar at a time, and it is generally easier to begin simplification by breaking the longest (uppermost) bar first. To illustrate, let's take the following expression and reduce it using DeMorgan's Theorems:

$$\boldsymbol{Y} = \overline{(\boldsymbol{A} + \overline{\boldsymbol{BC}})}$$

Correct Method

Following the advice of breaking the longest (uppermost) bar first, we should begin by breaking the bar covering the entire expression as a first step:

$$Y = \left(\overline{A + \overline{BC}}\right)$$

Breaking longest bar (addition changes to multiplication)

$$Y = \overline{A}.\overline{\overline{B.C}}$$

Applying identity $\overline{\overline{A}} = A$ to $\overline{\overline{BC}}$

$$\boxed{Y = \overline{A}.B.C}$$

Correct Answer

Incorrect Method

Following is the incorrect method for applying De Morgan's Theorem:

$$Y = \left(\overline{A + \overline{BC}}\right)$$

Breaking longest bar between A & B; Breaking both bars between B and C

INCORRECT STEP!

$$Y = \overline{A}.\overline{\overline{B}}.\overline{\overline{C}}$$

Applying identity $\overline{\overline{A}} = A$ to $\overline{\overline{BC}}$

$$\boxed{Y = \overline{A}.B + C}$$

Incorrect Answer

BASIC BOOLEAN ALGEBRA IDENTITIES

Following are some Boolean identities which will be very useful during simplification of Logic expressions:

Additive	Multiplicative
$A + 0 = A$	$0A = 0$
$A + 1 = 1$	$1A = A$
$A + A = A$	$AA = A$
$A + \overline{A} = 1$	$A\overline{A} = 0$

Fig. 3.2: Basic Boolean algebraic identities

LAB TASK 1

Simplify the following Boolean expression using De Morgan's theorem and Logic identities (attach working on a separate sheet):

$$Y = \overline{(A + B).(A + \overline{C})}$$

Logic Gate implementation of the original expression:

Input A	**Input B**	**Input C**	**Output Y**

Table 3.1: Truth table for the original expression

SIMPLIFIED EXPRESSION:

Logic Gate implementation for the simplified expression:

Input A	**Input B**	**Input C**	**Output Y**

Table 3.2: Truth table for simplified circuit

HARDWARE REQUIRED:

- Power supply with cables
- Breadboard
- Logic gate ICs: 1) AND Gate 2) OR Gate 3) NOT Gate
- LEDs
- Logic Probe (optional)

PROCEDURE:

1. First of all, draw the logic gate circuit of the original equation and build the Truth table for the circuit. (attach working on separate sheet)
2. Then simplify the expression using Boolean algebra identities and De Morgan Theorem. And then develop the truth table for the simplified circuit. (Attach working on separate sheet).
3. Now build the logic circuits for both original and simplified Boolean expression and complete the tables 3.1 and 3.2
4. Compare the two truth tables.
5. Does the theoretical calculations confirmed by Experimental findings? ________
6. List at least two advantages of simplifying Boolean expressions:

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LAB TASK 2

1. Simplify the following Boolean expression using De Morgan's theorem and Logic identities, if applicable. (attach circuit diagram / working on a separate sheet):

$$Y = \bar{A}BC + A\bar{B}C + AB\bar{C}$$

2. For the given Truth Table, realize a logical circuit using **basic gates** and **NAND** gates

Inputs				Output
A	**B**	**C**	**D**	**Y**
0	0	0	0	**1**
0	0	0	1	**1**
0	0	1	0	**0**
0	0	1	1	**0**
0	1	0	0	**1**
0	1	0	1	**1**
0	1	1	0	**0**
0	1	1	1	**0**
1	0	0	0	**0**
1	0	0	1	**0**
1	0	1	0	**0**
1	0	1	1	**0**
1	1	0	0	**0**
1	1	0	1	**1**
1	1	1	0	**0**
1	1	1	1	**1**

CONCLUSION (What have you learnt from this experiment?):

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LEARNING OUTCOMES

Upon successful completion of the lab, students will be able to:

LO1: Understand and validate De Morgan's Theorem.

LO2: Observe the advantages of simplifying a complex Boolean expression in Logic circuit designing.